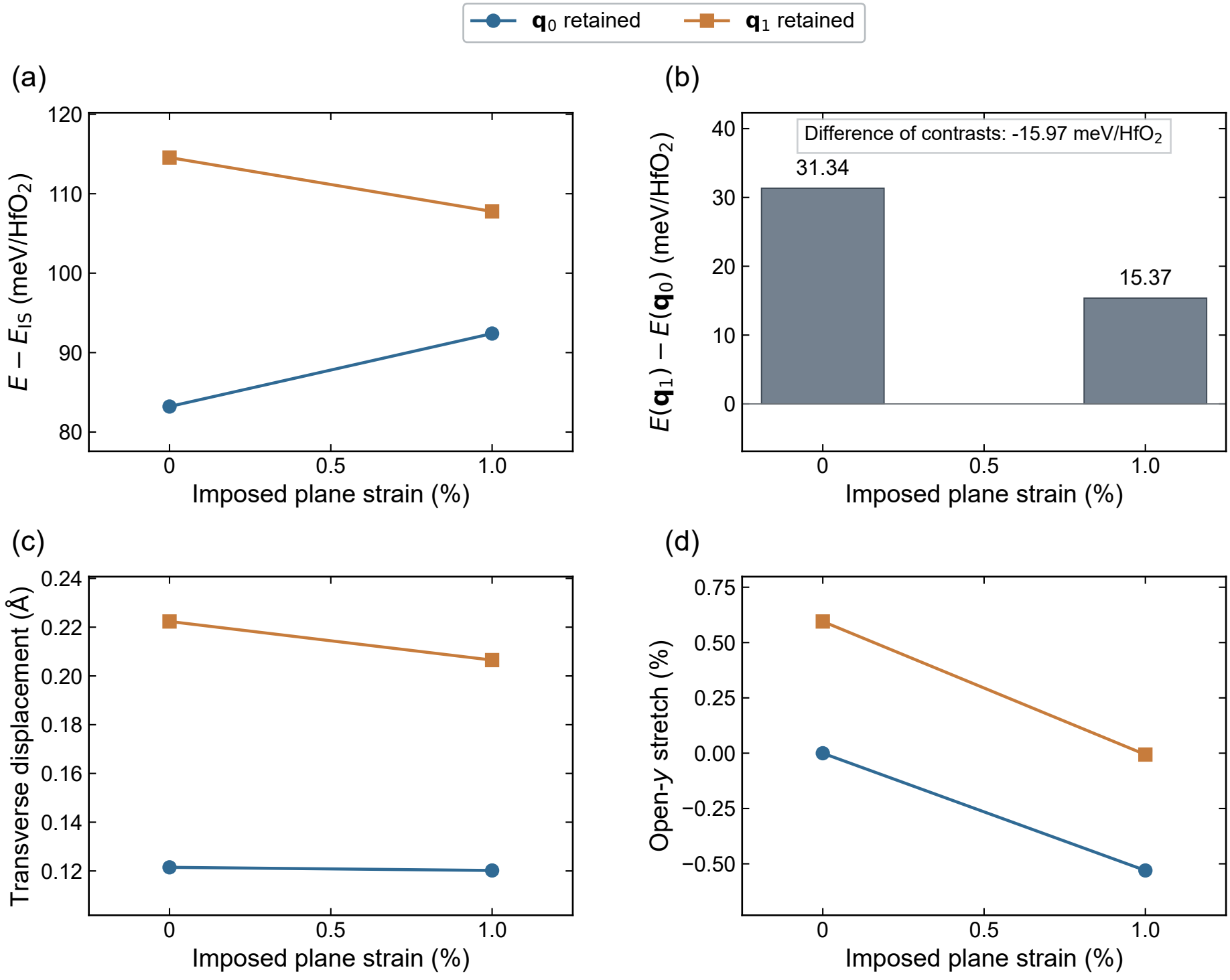
$\mathbf{q}_0$ retained
$\mathbf{q}_1$ retained
(a)
$E - E_{IS}$ (meV/HfO$_2$)
120
110
100
90
80
0
0.5
1.0
Imposed plane strain (%)
(b)
$E(\mathbf{q}_1) - E(\mathbf{q}_0)$ (meV/HfO$_2$)
Difference of contrasts: -15.97 meV/HfO$_2$
31.34
15.37
40
30
20
10
0
0
0.5
1.0
Imposed plane strain (%)
(c)
Transverse displacement (Å)
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0
0.5
1.0
Imposed plane strain (%)
(d)
Open-y stretch (%)
0.75
0.50
0.25
0.00
−0.25
−0.50
0
0.5
1.0
Imposed plane strain (%)